Transistores bipolares:

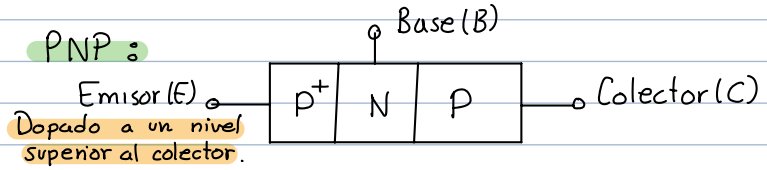
- Aspectos generales:

- Estructura interna:

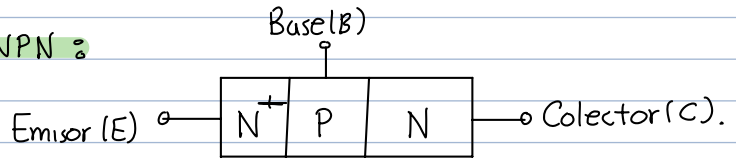
Dispositivo de 3 terminales basado en semiconductor que se usa en aplicaciones de amplificación y conmutación.

Formado por 2 uniones pn ⟶ Hay dos posibilidades npn o pnp.

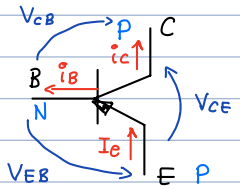
PNP:



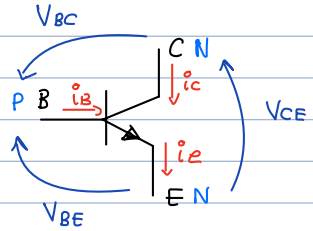
NPN:



Simbología: PNP ⇒



NPN ⟶



Para definir tensiones y corrientes ⟶ Transistor PNP, definimos tensiones siempre entre el lado P y el lado N.

Definiendo tensiones entre el lado P y el lado N.

Si teniendo tensiones entre $PN > 0$ ⟶ Transistor en directa

$PN < 0$ ⟶ Transistor en inversa.

También conocemos la tensión encerrada en la malla es igual a 0.

PNP ⟶ $V_{CE} = V_{CB} - V_{EB}$; $I_e = I_B + I_C$

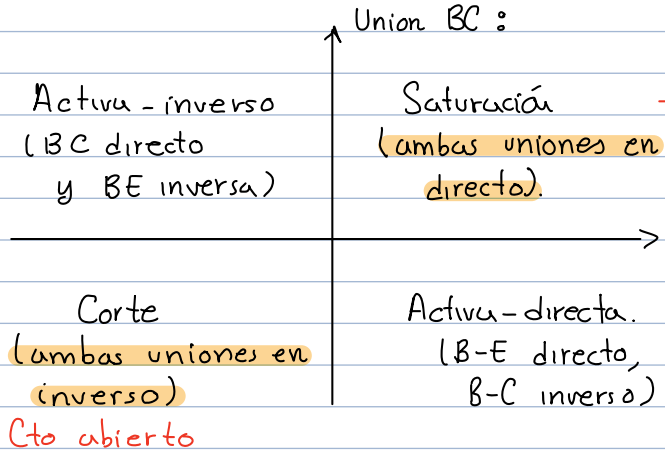
NPN ⟶ $V_{CE} = V_{BE} - V_{BC}$; $I_e = I_B + I_C$.

- Modos de funcionamiento:

Un transistor bipolar tiene 4 modos de funcionamiento

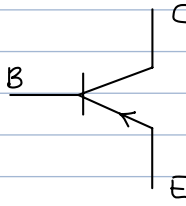
Corte: Aplicaciones digitales (como un interruptor abierto); Saturación: Aplicaciones digitales (como un interruptor cerrado)

Activa directa: Aplicaciones analógicas ; **Activa inversa**: muy poco utilizada.



→ Entre C.E pila de valor muy pequeño.

- Análisis cualitativo. Región activa directa.

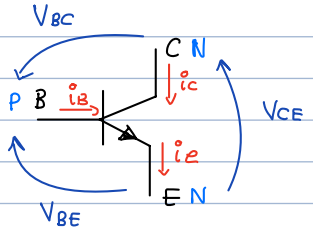


$V_{EB} > 0 \longrightarrow$ Unión BE en directa

$V_{CB} > 0 \longrightarrow$ Unión BC en inversa

$I_B > 0,\ I_C > 0,\ I_e > 0$

- Ecuaciones de Ebers Moll.



$$I_B - I_E - I_C = \frac{I_S}{\beta_F} \cdot \left(e^{\frac{V_{BE}}{KT/q}} - 1 \right) + \frac{I_S}{\beta_R} \cdot \left(e^{\frac{V_{BC}}{KT/q}} - 1 \right)$$

$$I_C = I_S \cdot \left(e^{\frac{V_{BE}}{KT/q}} - 1 \right) - \frac{(\beta_R + 1)}{\beta_R} \cdot I_S \cdot \left(e^{\frac{V_{BC}}{KT/q}} - 1 \right)$$

Se tratan de ecuaciones no lineales, relación v-i. Para ello aproximamos estas funciones no lineales por ecuaciones lineales.

Para un transistor NPN:

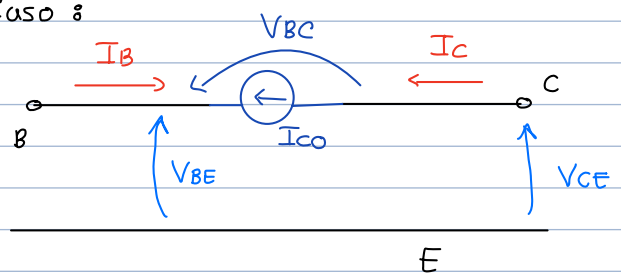
Zona de corte $\longrightarrow$ $\begin{cases} V_{BE} \leq V_{BE\gamma} \\ V_{BC} \leq V_{BC\gamma} \end{cases} \longrightarrow$

$I_C \simeq -I_B \simeq I_{CO}$ 1.

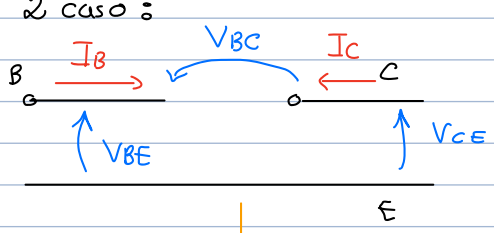
$I_C \simeq -I_B \simeq 0$ 2.

$I_e = 0.$

1º Caso:



2º caso:



↓ Suposición, se debe cumplir:

$$\begin{cases} V_{BE} \leq V_{BE\gamma} \\ V_{BC} \leq V_{BC\gamma} \end{cases}$$

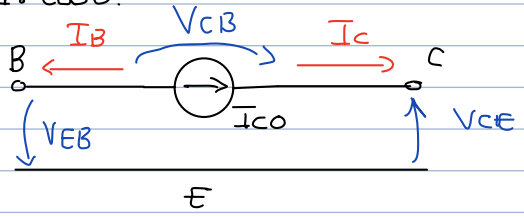
Para un transistor PNP

Corte $\begin{cases} V_{CB} \leq V_{CB\gamma} \\ V_{EB} \leq V_{EB\gamma} \end{cases} \longrightarrow$ $I_C \simeq -I_B = I_{CO}$ 1.

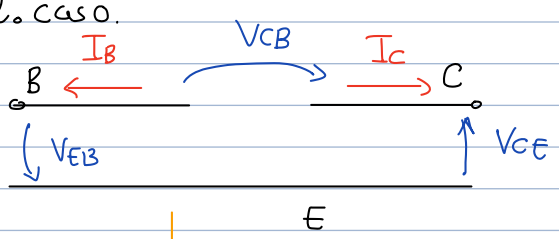
$I_C \simeq -I_B = 0$ 2.

$I_e = 0.$

1. caso.



2. caso.



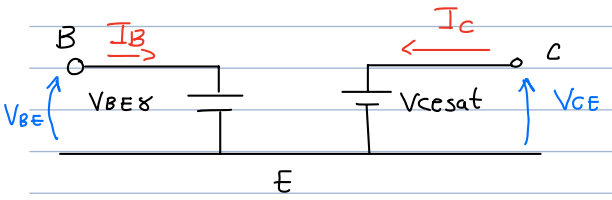
↓ Suposición, se debe cumplir.

$$\begin{cases} V_{EB} \leq V_{EB\gamma} \\ V_{CE} \leq V_{CE\gamma} \end{cases}$$

- Transistor NPN:

Saturación: $\begin{cases} V_{BE} = V_{BE\gamma} \rightarrow 0{,}6V \\ V_{BC} = V_{BC\gamma} \rightarrow 0{,}4V \end{cases} \longrightarrow \begin{cases} I_B \geq 0 \longrightarrow 1mA \\ I_C \geq 0 \longrightarrow 10mA \\ I_C \leq \beta_F I_B \longrightarrow Si\ \beta_F = 100 \longrightarrow Se\ cumple. \end{cases}$

La $V_{CE} = V_{BE} - V_{BC} = \underbrace{V_{BE\gamma} - V_{BC\gamma}}_{V_{CE\,sat}} = 0{,}2V.$



Tenemos que comprobar:

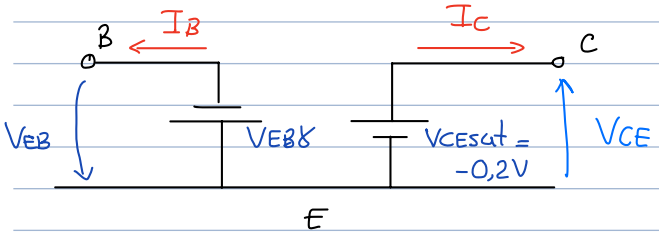
$$\begin{cases} I_B \geq 0 \\ I_C \geq 0 \\ I_C \leq \beta I_B \end{cases}$$

- Transistor PNP:

Saturación:

$$\begin{cases} V_{CB} = V_{CB\gamma} \simeq 0{,}4\,V \\ V_{EB} = V_{EB\gamma} \simeq 0{,}6\,V \end{cases} \longrightarrow \begin{cases} I_B \geq 0 \\ I_C \geq 0 \\ I_C \leq \beta_F I_B \end{cases}$$

$$\text{La } V_{CE} = V_{CB} - V_{EB} = \underbrace{V_{CB\gamma} - V_{EB\gamma}}_{V_{CEsat} < 0} \simeq 0{,}2 - 0{,}6 = -0{,}4\,V$$



Suponemos:

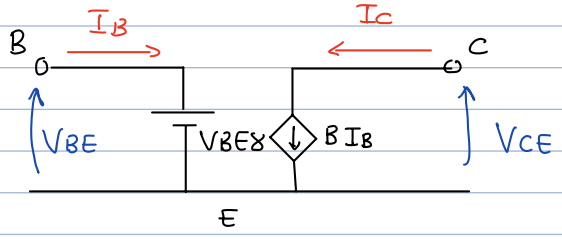
$$\begin{cases} I_B \geq 0 \\ I_C \geq 0 \\ I_C \leq \beta I_B \end{cases}$$

- Transistor NPN:

Activa:

$$\begin{cases} V_{BE} = V_{BE\gamma} \simeq 0{,}6\,V \\ V_{BC} \leq V_{BC\gamma} \simeq 0{,}4\,V \end{cases} \longrightarrow \begin{array}{l} I_B \geq 0 \\ I_C = \beta I_B + (\beta + 1) I_{CO} = \beta I_B \end{array}$$

$$V_{CE} = V_{BE} - V_{BC} = V_{BE\gamma} - V_{BC} \longrightarrow \begin{array}{l} \text{Si } V_{BC} = V_{BC\gamma} \Rightarrow V_{CE} = V_{CEsat}. \\ \text{Si } V_{BC} < V_{BC\gamma} \Rightarrow V_{CE} > V_{CEsat}. \end{array}$$



Suponemos:

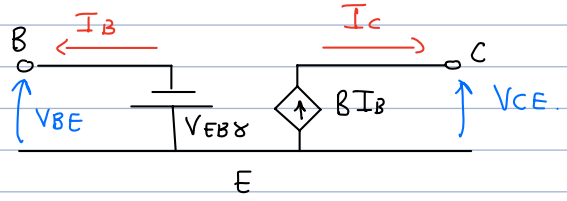
$$\begin{cases} I_B \geq 0 \\ V_{CE} \geq V_{cesat}. \end{cases}$$

- Transistor PNP :

Activa : $\begin{cases} V_{EB} = V_{EB\gamma} \simeq 0{,}6V & I_B \geq 0 \\ V_{CB} \leq V_{CB\gamma} \simeq 0{,}4V. & I_C = \beta I_B + (\beta+1)\cdot I_{CO} \simeq \beta \cdot I_B. \end{cases}$

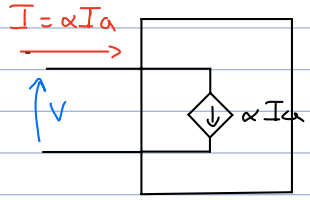
$V_{CE} = V_{CB} - V_{EB} = V_{CB} - V_{EB\gamma}$

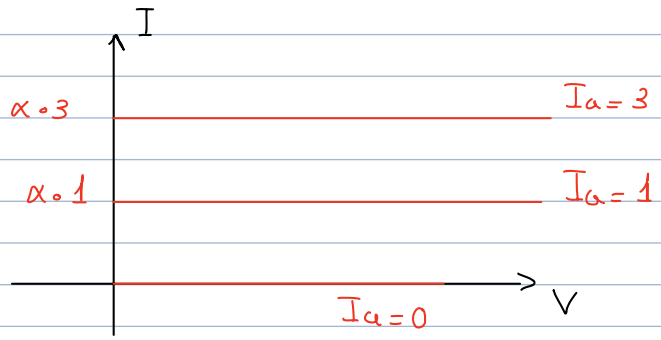
Un transistor PNP $\longrightarrow$ En activa sí $\longrightarrow$ $V_{CE} \leq V_{CE\,sat}$.

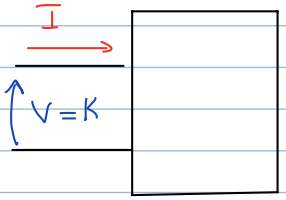


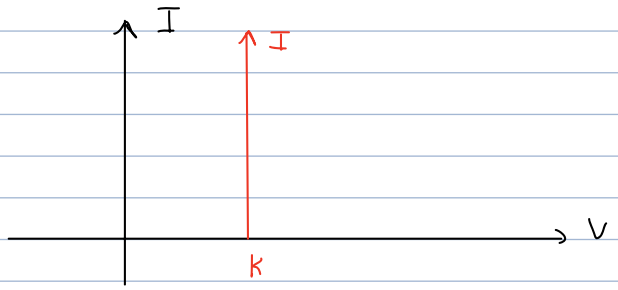
Suponemos :
- $I_B \geq 0$
- $V_{CE} \leq V_{CE\,sat}$.

- Curvas de entrada-salida :

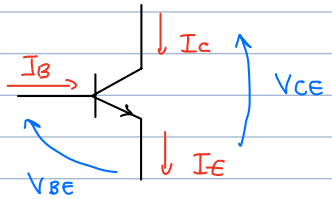




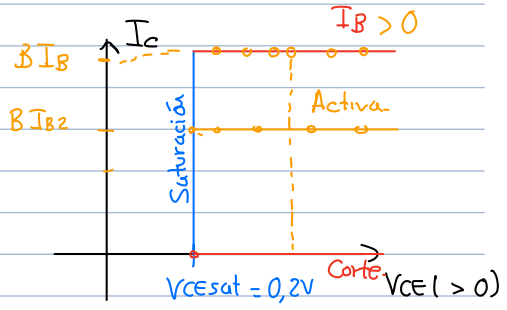




- NPN :



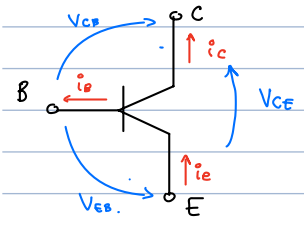


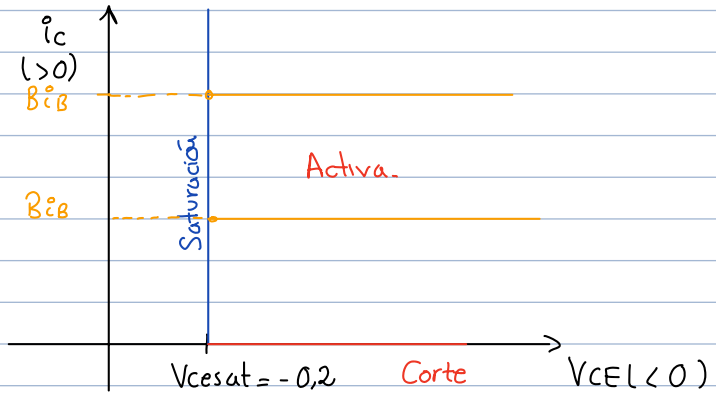


Saturación : $\begin{cases} I_B \geq 0 \\ I_C \geq 0 \\ \beta I_B \geq I_C \end{cases}$

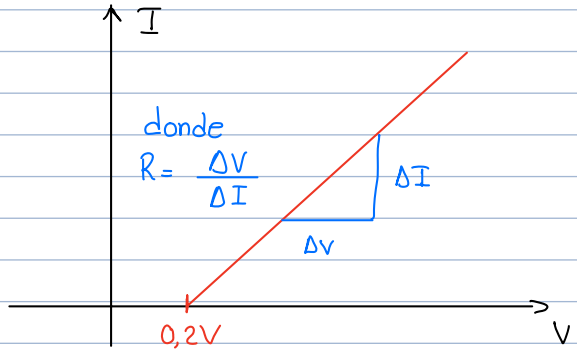
Activa : $\begin{cases} I_B \geq 0 \\ I_C = \beta I_B \\ V_{CE} \geq V_{CE\,sat} \end{cases}$

- PNP.

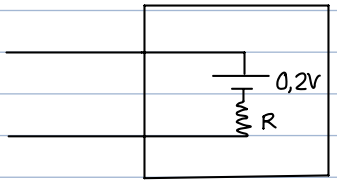




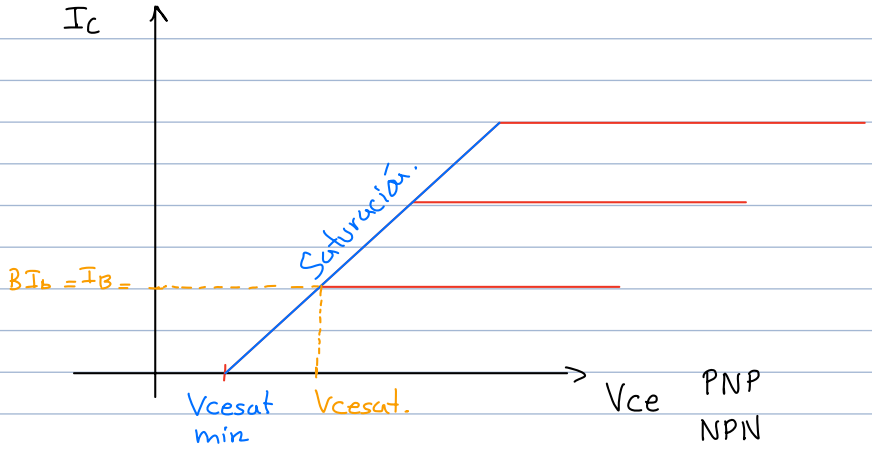
- Efectos de segundo orden:



$\Rightarrow$

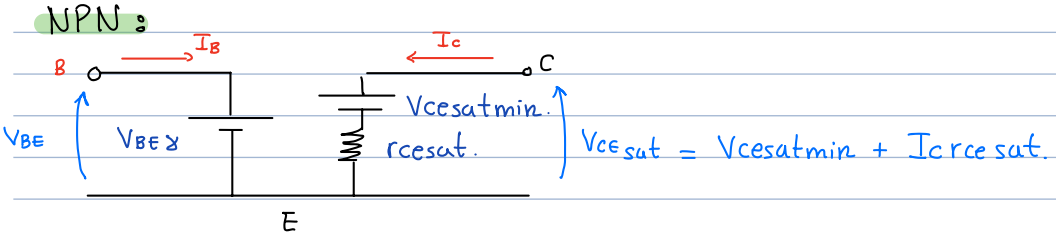


Transistores NPN y PNP:

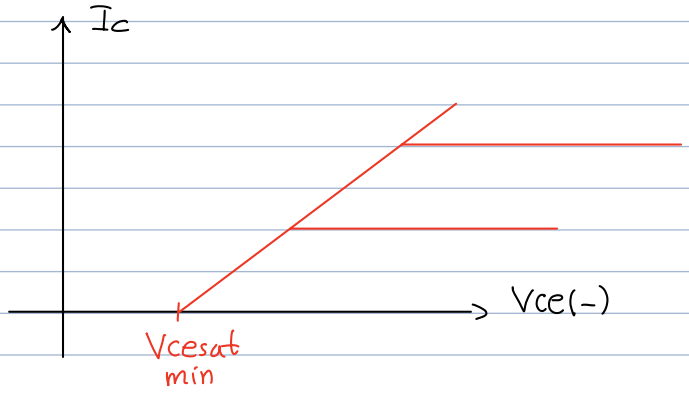


- Saturación:

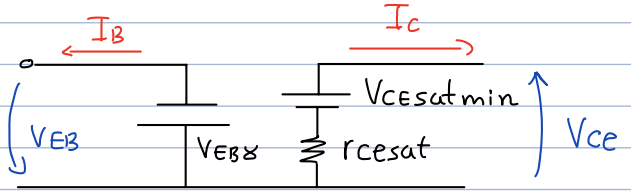
NPN:



PNP:



- Saturación:



$V_{cesat} = V_{CEsat\,min} - I_c r_{cesat}$.

- Análisis en DC:

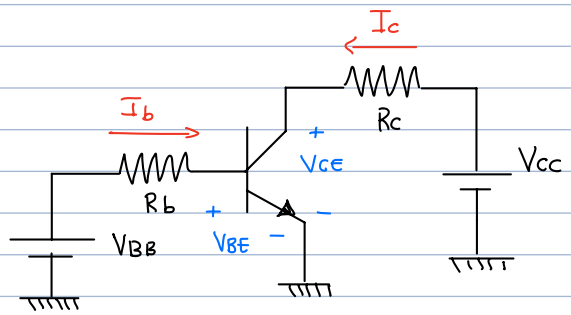
1. Hipótesis zona de trabajo del BJT
2. Sustituir BJT por su modelo equivalente
3. Resolver sistema de ecuaciones.

Zona activa: a comprobar →
- npn: $I_b > 0$, $V_{CE} \geq V_{CEsat}$.
- pnp: $I_b > 0$, $V_{ce} \leq V_{cesat}$

Zona saturación: a comprobar → $I_b > 0$, $I_c \leq B \cdot I_b$.

Zona corte: a comprobar →
- npn: $V_{BE} \leq V_\gamma$, $V_{BC} \leq V_\gamma$
- pnp: $V_{EB} \leq V_\gamma$, $V_{CB} \leq V_\gamma$

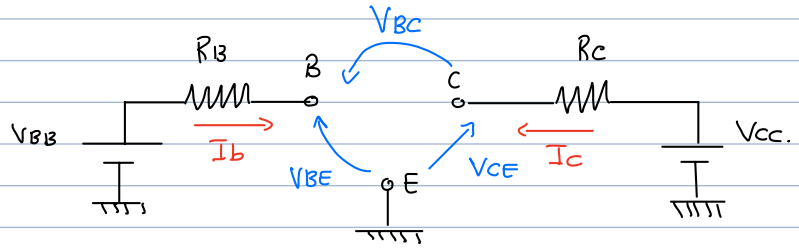
Ej:



Datos:
- $V_{BE\gamma} = 0{,}6V$
- $V_{CE\,sat} = 0{,}2V$
- $B_F = 200$
- $V_{BB} = 5V$
- $V_{cc} = 10V$
- $R_B = 1K\Omega$; $R_c = 10K\Omega$.

- Suposiciones :

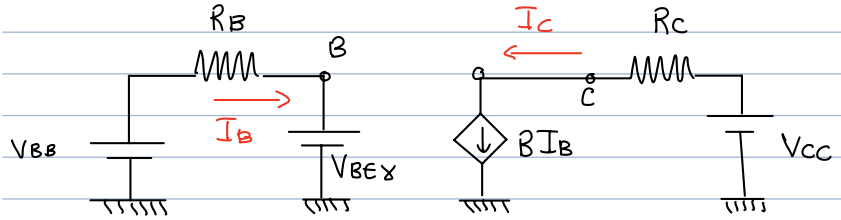
1. Suponemos en corte : ⟶ $\begin{cases} V_{BE} \leq V_\gamma \Rightarrow 5V \leq 0{,}6V \quad \text{No!} \\ V_{BC} \leq V_\gamma \end{cases}$



Donde $V_{BE} = 5V$.

Transistor no está en corte.

2. Suponemos en activa : ⟶ $\begin{cases} I_B \geq 0 \\ V_{CE} \geq V_{cesat} = 0{,}2V \end{cases}$

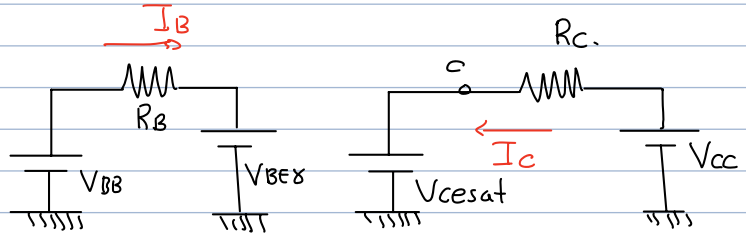


$I_B = \frac{V_{BB} - V_{BE\gamma}}{R_B} = \frac{5-0{,}6}{1K\Omega} = 4{,}4\ mA.$ $\quad \beta I_B = I_C \Rightarrow 200 \cdot 4{,}4\ mA$

$V_{CE} = -\beta I_B \cdot R_C + V_{CC} = -200 \cdot 4{,}4\ mA \cdot 10K\Omega + 10V \Rightarrow$ Negativo

$V_{ce} \geq V_{cesat} \longrightarrow$ No es mayor $\longrightarrow$ No está en activa.

3. Suponemos en saturación : ⟶ $I_b \geq 0$; $I_c < \beta \cdot I_B$; $I_c \geq 0$.

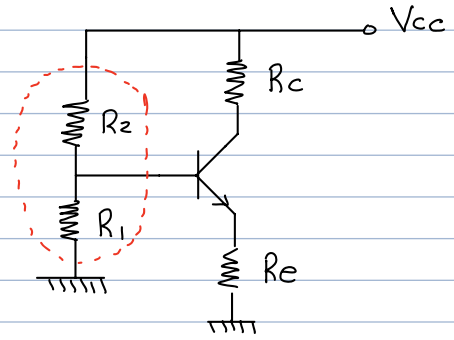


$I_B = \frac{V_{BB} - V_{BE\gamma}}{R_B} = 4{,}4\ mA$; $I_C = \frac{V_{CC} - V_{cesat}}{R_C} = \frac{10-0{,}2}{10K} = 0{,}98\ mA.$

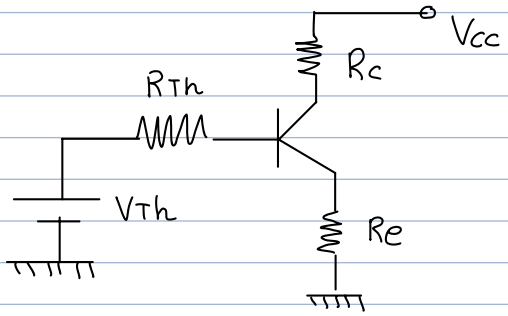
$I_B \geq 0$; Sí . $I_C \geq 0$; Sí $\quad 0{,}98\ mA \leq 200 \cdot 4{,}4\ mA \longrightarrow$ Sí .

El dispositivo se encuentra en saturación.

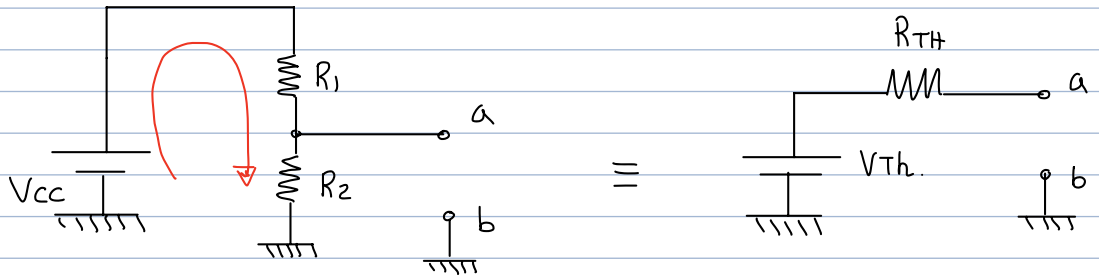
Ej:



Sustituir por equiv de Thevenin:



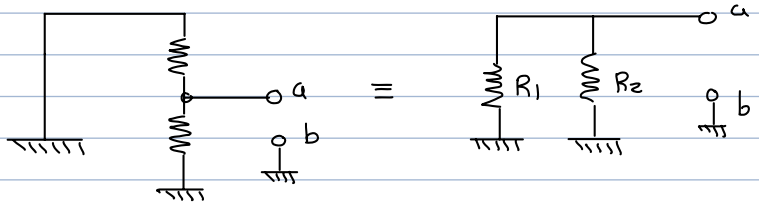
Para calcular el equivalente de Thevenin:



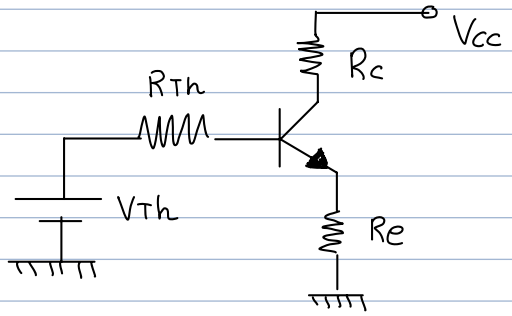
- Para el cálculo de VTh →

$$V_{Th} = \frac{V_{cc}}{R_1 + R_2} \cdot R_2$$

- Para el cálculo de RTh → Impedancia vista cuando anulamos la pila.



$$\Rightarrow R_{Th} = R_1 // R_2 = \frac{R_1 \cdot R_2}{R_1 + R_2}$$



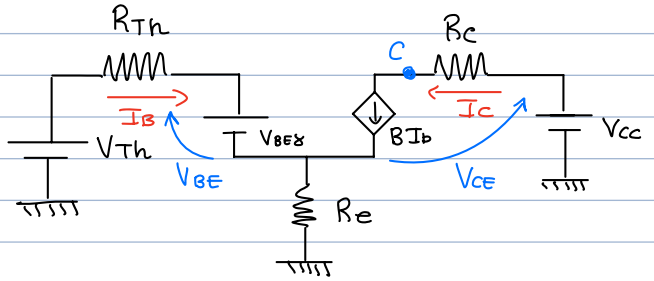
donde $V_{Th} = 5V$
$R_{Th} = 3,33\ K\Omega$
$V_{cesat} = 0,2V$
$V_{BE} = 0,7V$

$B \Rightarrow \begin{cases} B = 100 \\ B = 300 \end{cases}$

$V_{cc} = 15V.$
$R_e = 1K\Omega$
$R_c = 1K\Omega.$

Suposiciones :

1. Activa $\longrightarrow \begin{cases} I_C \geq 0 \\ V_{CE} \geq V_{CEsat} \end{cases}$



Analizamos cto :

$$-V_{Th} + R_{Th} \cdot I_b + V_{BE\gamma} + R_e(B+1)I_b = 0$$

$$I_b = \frac{V_{Th} - V_{BE\gamma}}{R_{Th} + (B+1)R_e}$$

Para $B = 100 \Rightarrow I_b = \frac{5 - 0{,}7V}{3{,}33K + (101) \cdot 1K\Omega} = 41{,}2\,\mu A > 0$

Para $B = 300 \Rightarrow I_b = \frac{5 - 0{,}7}{3{,}33 + (301) \cdot 1K} = 14{,}1\,\mu A > 0.$

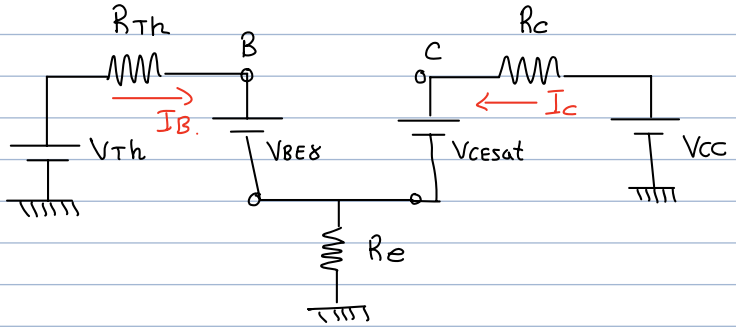
Calculamos $V_{CE} \Rightarrow -BI_b \cdot R_c + V_{cc} - R_e(B+1)I_b =$

Para $B = 100 \Rightarrow -100 \cdot 41{,}2\mu \cdot 1K + 15V - 1K \cdot 101 \cdot 41{,}2\mu A = 6{,}72\,V$

$6{,}72\,V \geq 0{,}2\,V \Rightarrow$ Sí

El transistor está en zona activa

2. Saturación $\longrightarrow \begin{cases} I_B \geq 0 \\ I_C \geq 0 \\ BI_b \geq I_C \end{cases}$



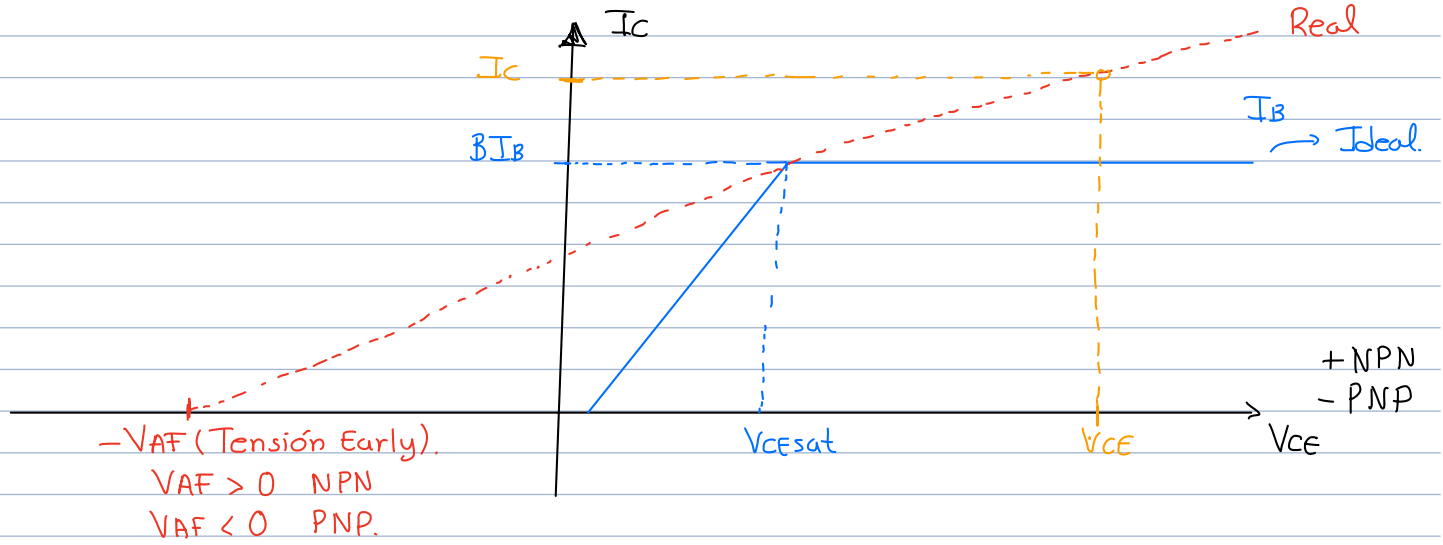
$$-V_{Th} + R_{Th} \cdot I_b + V_{BE\gamma} + R_e(I_b + I_c) = 0.$$

$$-V_{cc} + R_e \cdot I_c + V_{CEsat} + R_e(I_b + I_c) = 0.$$

Sabemos que no esta en saturación, pero este sería el procedimiento.

- Efecto Early :

Curvas de salida de un transistor :



$$\frac{V_{CE}+V_{AF}}{I_C} = \frac{V_{CEsat}+V_{AF}}{\beta I_B} \simeq \frac{V_{AF}}{\beta I_B} \Rightarrow I_C = \beta I_B \left[\frac{V_{CE}+V_{AF}}{V_{AF}}\right]$$

$$I_C = \beta I_B + \frac{V_{CE}}{V_{AF}} \cdot \beta I_B = \beta I_B + \frac{V_{CE}}{r_o} \; ; \; r_o = \frac{V_{AF}}{\beta I_B}$$

$$I_C = \beta I_B + \frac{V_{CE}}{(V_{AF}/\beta I_B)} = \longrightarrow$$

1. NPN (Activa)

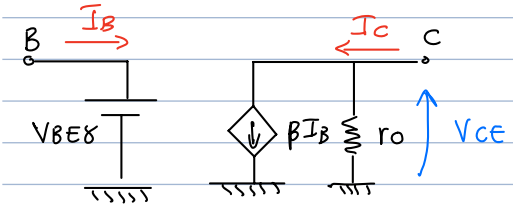
$$I_C = \beta I_B + \frac{V_{CE}}{(V_{AF}/\beta I_B)} = \beta I_B + \frac{V_{CE}}{r_o}$$

$$r_o = \frac{|V_{AF}|}{\beta I_B}$$

2. PNP. (Activa)

$$I_C = \beta I_B - \frac{V_{CE}}{(-V_{AB}/\beta I_B)} = \beta I_B - \frac{V_{CE}}{r_o}$$

1. NPN.



2. PNP.

